

AngularJS – 40 Hours Training Plan

Module	Topics Covered	Hours	Learning Outcomes
1. Introduction to AngularJS	- What is AngularJS? - Features & benefits - Setting up development environment	2	Understand AngularJS basics and development setup
2. AngularJS Architecture	- MVC pattern - Directives, Controllers, and Scope - Data Binding	3	Learn the architecture and how different components interact
3. Expressions and Data Binding	- Expressions in HTML - Two-way data binding - ng-bind vs interpolation	3	Work with dynamic data and bind it to UI
4. Controllers and Scope	- Defining controllers - Using \$scope object - Scope inheritance	3	Structure application logic using controllers
5. Directives in AngularJS	- Built-in directives (ng-model, ng-repeat, etc.) - Custom directives - Directive restrictions	4	Use and create custom behavior using directives
6. Filters and Services	- Built-in filters (currency, date, filter) - Creating custom filters - Using services (built-in and custom)	4	Format data and manage reusable logic
7. Forms and Validation	- Form controls & ng-model - Validating forms - Custom validations	4	Build and validate forms using AngularJS
8. Dependency Injection	- What is DI? - Injecting services into controllers - Factory, service, and provider	3	Manage dependencies effectively across modules
9. Routing and SPA Concepts	- ngRoute module - Single Page Application (SPA) - Route parameters and templates	4	Create dynamic single-page applications with routing
10. AJAX and RESTful Integration	- \$http service - Fetching data from APIs - CRUD operations using REST	4	Connect AngularJS to backend services using HTTP
11. Project Work	- Mini project using all concepts - Build a basic CRUD app (e.g., Student Management)	6	Apply all skills in a hands-on real-world application
12. Best Practices and Optimization	- Code organization - Performance tips - Debugging tools	2	Write clean, maintainable, and performant code